APPLIED STATISTICS, MATH10096

Workshop 2

1. Consider the linear model
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$
where $\mathbf{y}$ is an $n \times 1$ vector of response variables, $\mathbf{X}$ is an $n \times p$ design matrix $(p < n)$ of full column rank, and $\boldsymbol{\beta}$ is a $p \times 1$ parameter vector. Further, $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of independent, zero mean, normal random variables, all with the same variance, σ^2. The QR decomposition of the design matrix is
$$\mathbf{X} = \mathbf{Q}\begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix},$$
where $\mathbf{Q}$ is an $n \times n$ orthogonal matrix, $\mathbf{R}$ is a $p \times p$ upper triangular matrix, and $\mathbf{0}$ is a $(n-p) \times p$ matrix of zeroes. Furthermore, let $\mathbf{Q} = (\mathbf{Q}_1, \mathbf{Q}_2)$, where $\mathbf{Q}_1$ is $n \times p$ and formed by the first p columns of $\mathbf{Q}$, whereas $\mathbf{Q}_2$ is formed by the remaining $n-p$ columns of $\mathbf{Q}$.

   (a) Show that the least squares estimator of $\boldsymbol{\beta}$ is
$$\widehat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{Q}_1^T\mathbf{y}.$$

   (b) Derive the distribution of the least squares estimator, $\widehat{\boldsymbol{\beta}}$, in terms of $\boldsymbol{\beta}$, $\mathbf{R}$, and σ^2, and hence derive the distribution of $(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}$, where $\sigma_{\hat{\beta}_j}$ is the standard deviation of $\hat{\beta}_j$, for $j = 1, \ldots, p$.

   (c) Show that
$$\mathbf{Q}_2^T\mathbf{y} \sim \mathrm{N}(\mathbf{0}_{n-p}, \sigma^2\mathbf{I}_{n-p}).$$

   (d) Derive the distribution of $\|\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\|^2/\sigma^2$.

2. In the footnote of Page 39 of the lecture notes, it was stated that the orthogonality between the fitted values $\hat{\boldsymbol{\mu}}$ and the residuals $\hat{\boldsymbol{\epsilon}}$ implies, provided that the model includes an intercept, that the sample correlation between $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\epsilon}}$ must be zero. Verify this claim by computing the sample correlation between $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\epsilon}}$.

3. Suppose that you fit a linear model to some response data, $\mathbf{y}$ and obtain residuals, $\hat{\boldsymbol{\epsilon}}$. Now suppose that you fit exactly the same model but with this $\hat{\boldsymbol{\epsilon}}$ vector treated as the response. What would the fitted value vector be for the second model fit?

4. Consider some data for deformation (in mm), y_i, of 3 different types of alloy, under different loads (in kg), x_i. When there is no load, there is no deformation, and the deformation is expected to vary linearly with load, in exactly the same way for all three alloys. However, as the load increases the three alloys deviate from this ideal linear behaviour in slightly different ways, with the relationship becoming slightly curved (possibly suggesting quadratic terms). The loads are known very precisely, so errors in x_i's can be ignored, whereas the deformations, y_i, are subject to larger measurement errors, that do need to be taken into account. Define a linear model suitable for describing these data, assuming that the same 6 loads are applied to each alloy, and write it out in the form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$.

5. Suppose that $\mathbf{y}$ is a response vector and $\mathbf{x}$, $\mathbf{v}$ and $\mathbf{z}$ are vectors of predictor variables. What is wrong with testing

$$\mathrm{H}_0 : y_i = \beta_1 + \beta_2 x_i + \beta_3 z_i + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2),$$

against

$$\mathrm{H}_1 : y_i = \beta_1 + \beta_2 v_i + \beta_3 z_i + \beta_4 z_i^2 + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2),$$

using the F-ratio results from Section 4.3.2 of the notes?

6. A statistician has fitted two alternative models to response data y_i. The first is

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{1}$$

and the second is

$$y_i = \beta_0 + \beta_1 x_i + \gamma_j + \epsilon_i \quad \text{if } y_i \text{ from group } j. \tag{2}$$

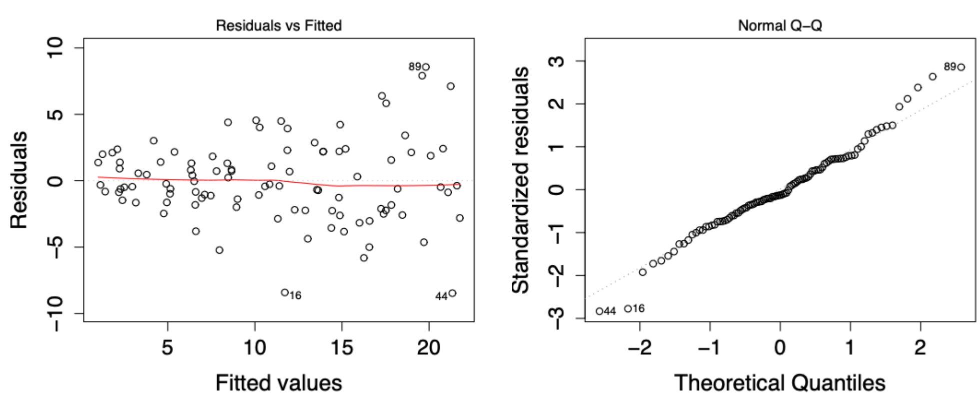
In `R` the factor variable containing the group labels is `trt`. The statistician wants to test the null hypothesis that model (1) is correct against the alternative that model (2) is correct. To do this, both models are fitted in R, and a fragment of the summary for each is shown below.

```
summary(b0)
lm(formula = y ~ x)
...
Residual standard error: 0.3009 on 98 degrees of freedom

summary(b1)
lm(formula = y ~ x + trt)
...
Residual standard error: 0.3031 on 95 degrees of freedom
```

In `R` this test could be conducted via `anova(b0,b1)`, but instead perform the test using just the information given.

7. Two statisticians disagree about the following pair of residual plots from a linear model.



Statistician A says that the left hand plot shows evidence of a mean variance relationship, violating the constant variance assumption, indicating a need to fix the model. Statistician B says that although the left hand plot looks problematic at first sight, the right hand Q-Q plot shows that the residuals are close enough to normality that the apparent mean-variance relationship will have no adverse effect. Who is right, and why?

8. A 'quant' working for a prestigious international financial firm proposes the following model for predicting the half hour return rate, r_i, on a particular short term trade, based on simple market index and volatility data:

$$r_i = \beta_0 + \beta_1 \cos(2t_i\pi/48) + \beta_2 \sin(2t_i\pi/48) + \beta_3 \texttt{DJ}_{i-1} + \beta_4 \texttt{DAX}_{i-1} + \beta_5 \texttt{FT}_{i-1} \\ + \beta_6(\texttt{DJ}_{i-1} - \texttt{DJ}_{i-2})^2 + \beta_7(\texttt{DAX}_{i-1} - \texttt{DAX}_{i-2})^2 + \beta_8(\texttt{FT}_{i-1} - \texttt{FT}_{i-2})^2 + \epsilon_i \quad (3)$$

where the ϵ_i are i.i.d. $N(0, \sigma^2)$, r_i is the return in the i^{th} half hour period, t_i is the time measured in half hours from some arbitrary start point, $\texttt{DJ}_i$, $\texttt{DAX}_i$ and $\texttt{FT}_i$ are the state of the Dow Jones, DAX and FT100 indices in the i^{th} half hour period. The $\sin$ and $\cos$ terms are there to model daily fluctuations. The model is to be estimated by least squares using data for the fortnight leading up to the trading half hour of interest. There is money to be made if r_i can be predicted along with accurate assessment of the prediction uncertainty. In test runs the prediction accuracy seems reasonable.

   What is likely to be wrong with this model, and what residual plot would you use to show the problem up?

9. A distillery sets up and sponsors a hill race dubbed the 'Whisky challenge', for promotional purposes. To generate extra interest from elite fell runners in the first year, it is proposed to offer a prize for every runner who completes the course in less than a set time, T_0. The organisers need to set T_0 high enough to generate a big field of participants, but low enough that they do not bankrupt the distillery. To this end they approach you to come up with a predicted winning time for the race. To help you do this, the `hills` data frame in R package `MASS`, provides winning times for 35 Scottish hill races. To load the data and examine it type `library(MASS);hills` in R. Find and estimate a suitable linear model for predicting winning `times` (in minutes) in terms of race distance `dist` (in miles) and the total height climbed `climb` (in feet). The 'Whisky Challenge' is to be a 7 mile race, with 2400 feet of ascent. Produce a short report for the race organisers. *Note*: It can be instructive to look at 'Naismith's rule' on Wikipedia.